ME 5620 Project

Charge and discharge of a solid storage block with internal electric heater and air as HTF

December 9, 2025

Luke Esperiquette

Why Thermal Energy Storage?

- Heavy industry uses ~30% of global greenhouse gas emissions for heat and power.
- Renewable electricity (solar/wind) is cheap but intermittent.
- Electro-thermal TES: convert electricity → heat → store in solid → later deliver hot air or power.
- This project models **one small TES cell** similar in concept to Antora's carbon block systems.



Project Description & Deliverables

3D conjugate heat transfer model in ANSYS Fluent

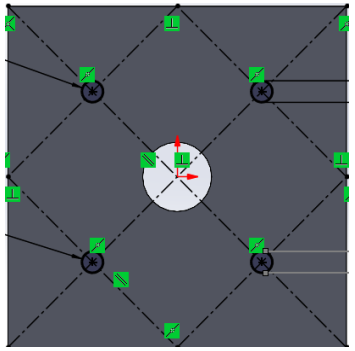
8-hour charge (heater on, no flow) + 2-hour discharge (air flow, no heater)

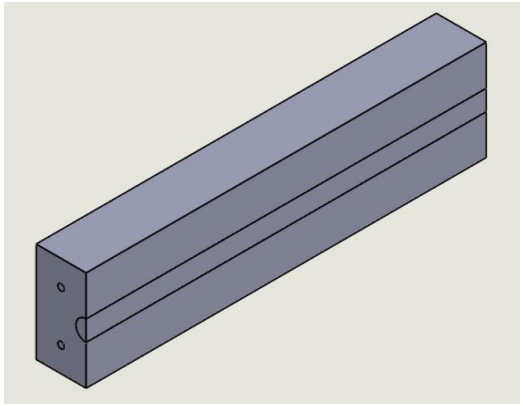
Required post-processing:

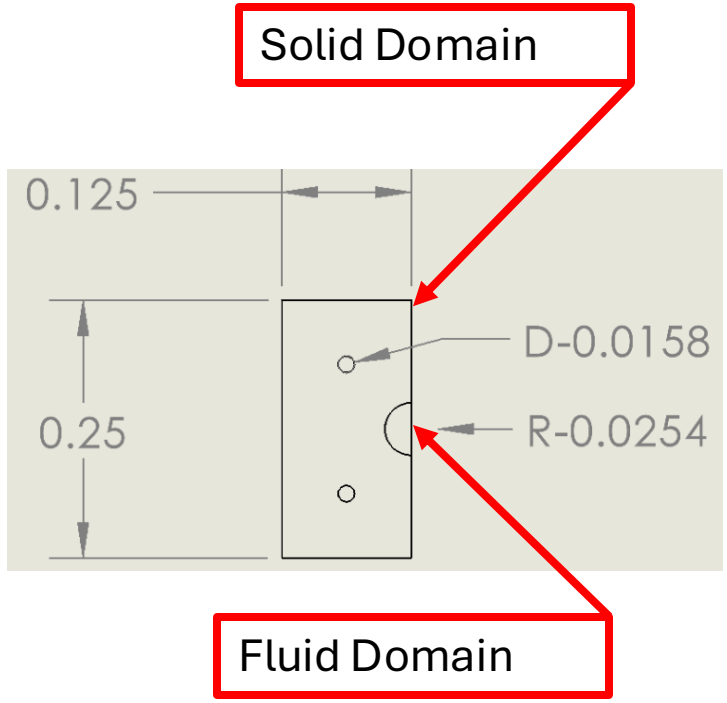
Volume-weighted average SM temperature	Mass-weighted average HTF exit temperature	Area-weighted surface heat flux between HTF and SM	Velocity & temperature contour plots and animation	Energy stored, energy removed, round-trip efficiency

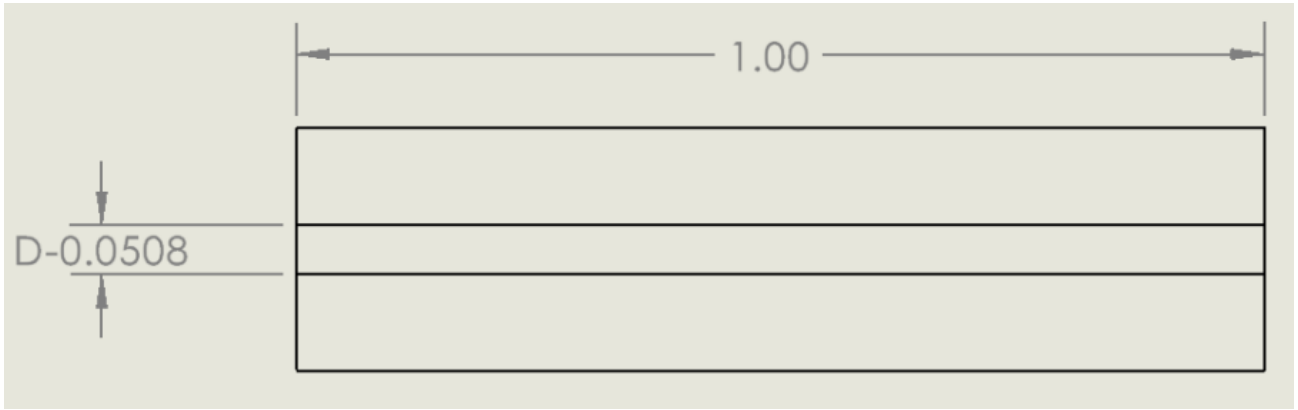
Geometry & Modeling

- Storage block: **0.25 m × 0.25 m × 1 m** solid (sensible storage material)
- Central air tube and cylindrical electric heater embedded in the block
- Perfect insulation on outer faces (no heat loss to environment)
- Air flows only through the tube during discharge
- ½ model of the geometry modeled with symmetry planes
- Units: m



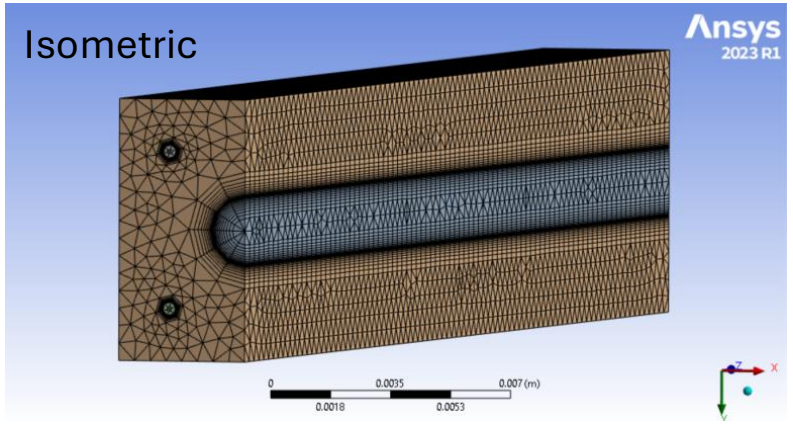


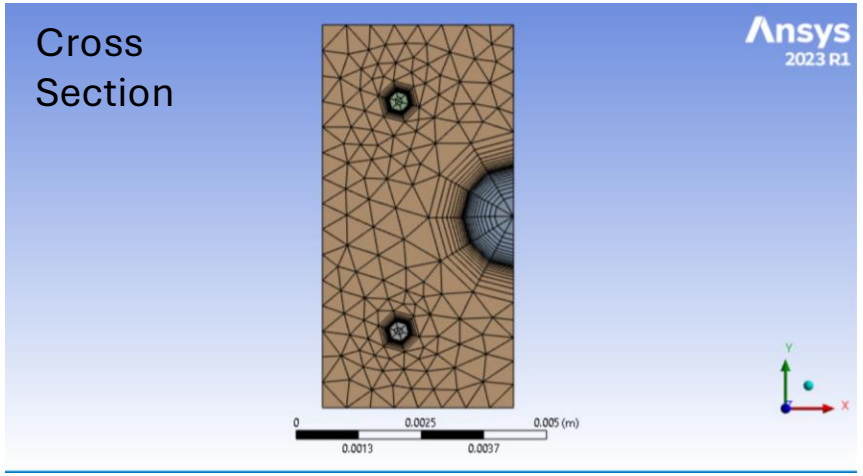




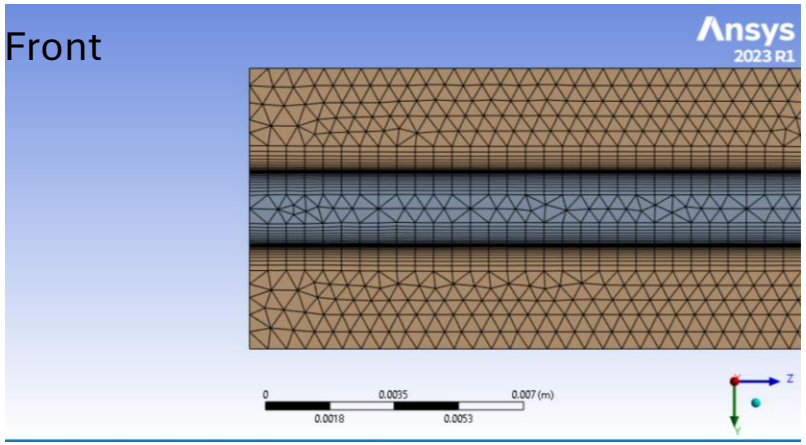
Mesh

Grid Refinement

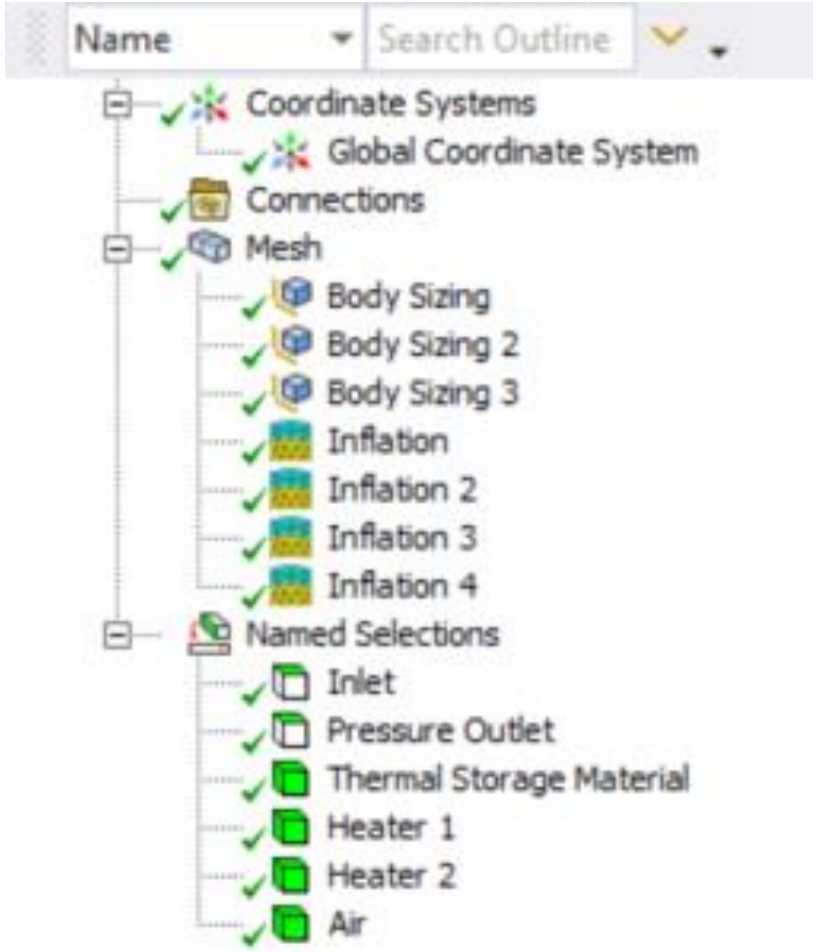




- Unstructured tetrahedral mesh with local refinement near:
  - air/solid interface
  - heater surface
  - Inflation layers in the tube to resolve near-wall gradients (for velocity and heat transfer).
- Grid-refinement study:
  - Coarse / medium / fine meshes
- Compared volume-avg SM temperature and HTF exit temperature at a fixed time



Mesh Settings



Storage Material Properties

- Storage material (SM)
  - ρ = 1300 kg/m^3
  - cp = 1200 J/kg·K
  - k = 0.4 W/m·K
- Heater:
  - stainless steel,
  - constant properties (ρ, cp, k from Fluent database)
- Heat Transfer Fluid (air):
  - ρ = 0.6746 kg/m^3
  - cp = 0.038 J/kg·K
  - k = 0.000025 W/m·K

- Mass flow rate during discharge: 0.03373 kg/s of air

Simulation Setup

- Transient simulation, total flow time = **10 hours**
  - Charge: 0–8 h (heater on, no flow)
  - Discharge: 8–10 h (flow on, heater off)

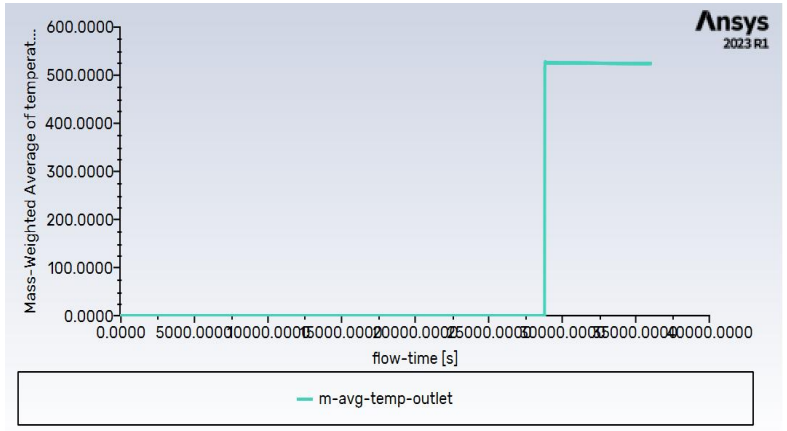
Charge:

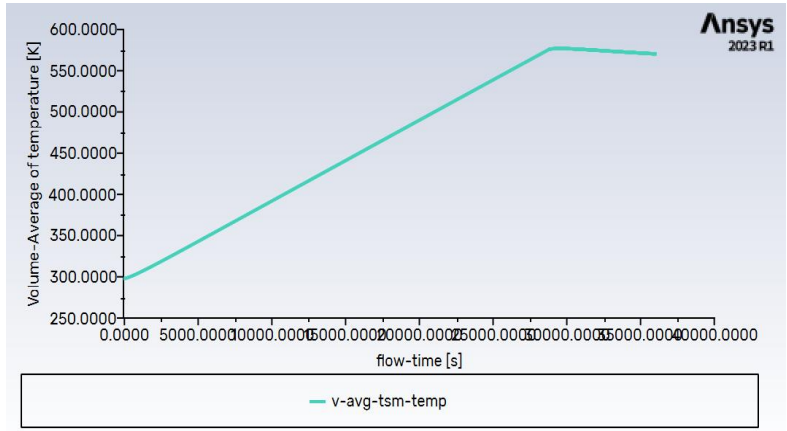
- Energy equation ON, flow/turbulence equations OFF
- Uniform heat generation in heater body, tuned to 523 °K

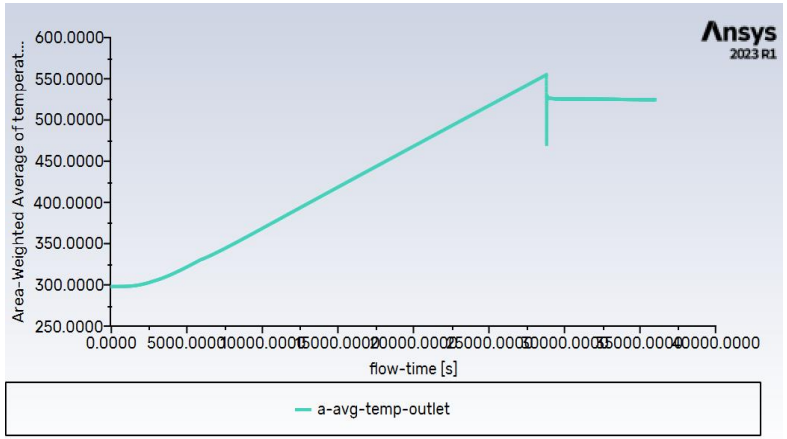
Discharge:

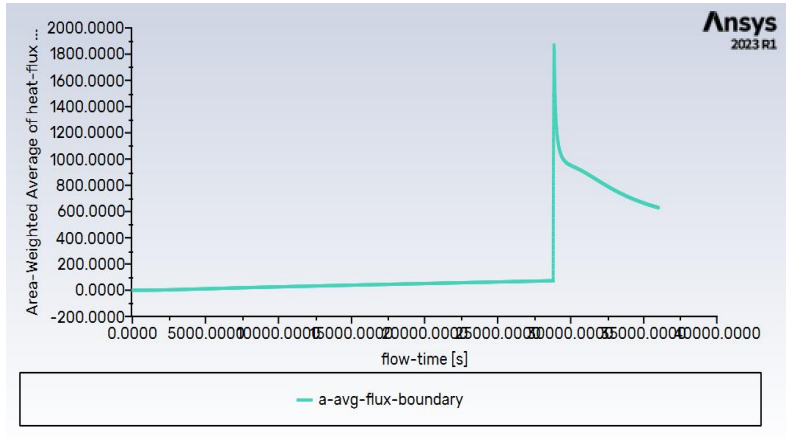
- Mass flow inlet (0.03373 kg/s) with Tin = 298 °K
- Pressure outlet at 1 atm
- Turbulence model: k-ω
- Outer walls adiabatic; no convection to ambient

Monitoring Quantities (Reports) Mesh #1 (fine)





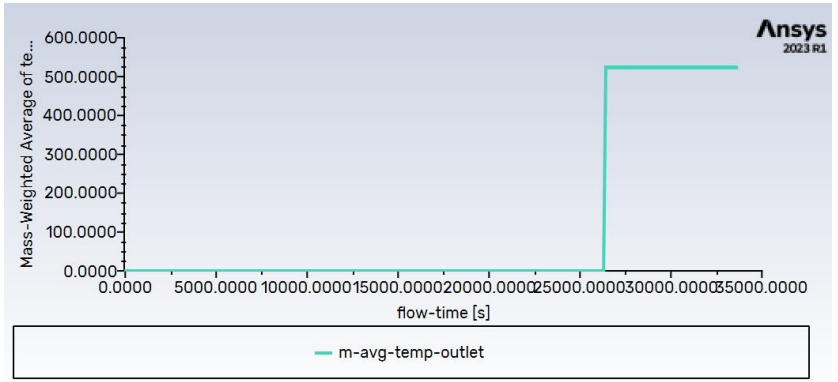


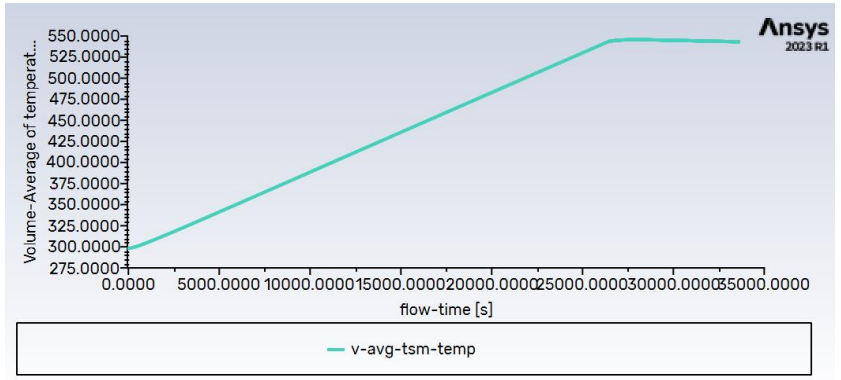


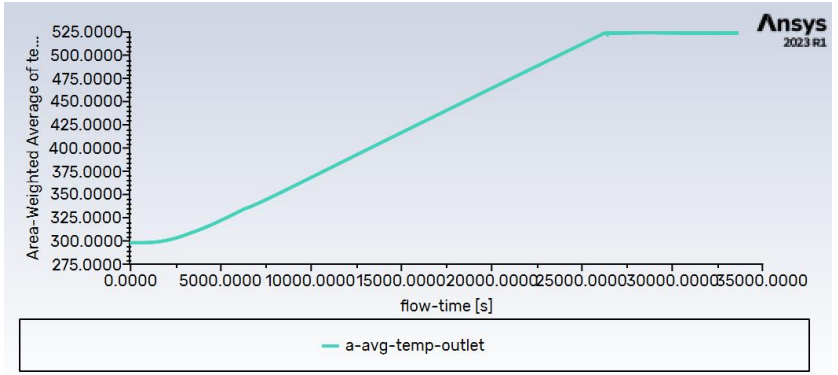
Fluent report definitions:

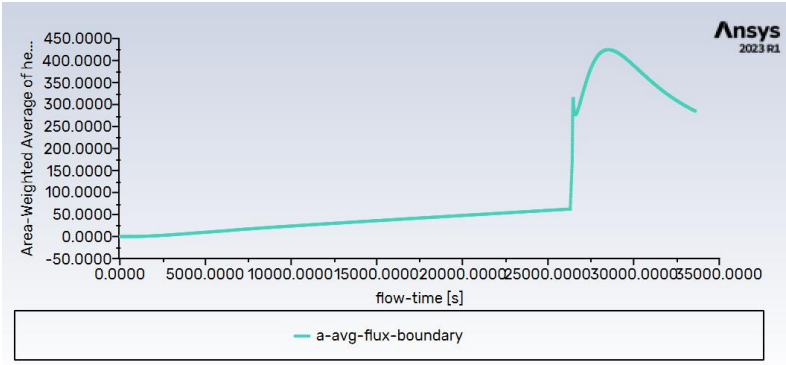
- Volume-weighted average **SM temperature** (storage block)
- Mass-weighted average **HTF exit temperature** (air outlet)
- Area-weighted **surface heat flux** between SM and HTF (tube wall)

Monitoring Quantities (Reports) Mesh #2 (mid)







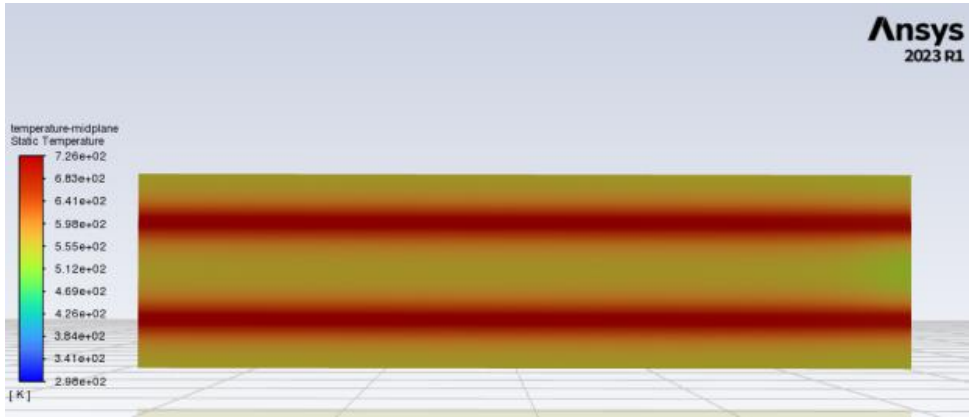


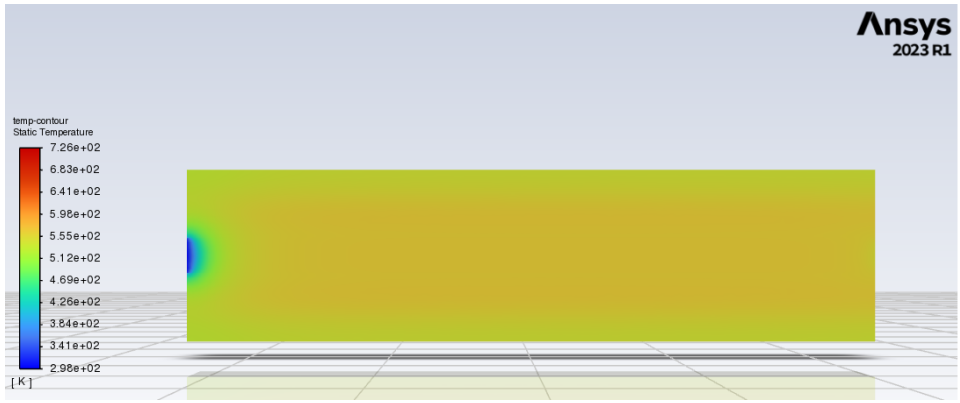
Fluent report definitions:

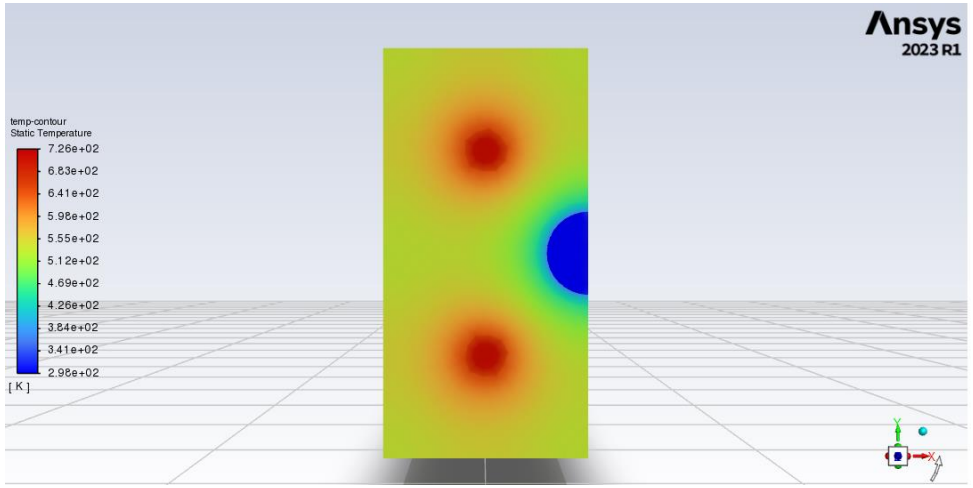
- Volume-weighted average **SM temperature** (storage block)
- Mass-weighted average **HTF exit temperature** (air outlet)
- Area-weighted **surface heat flux** between SM and HTF (tube wall)

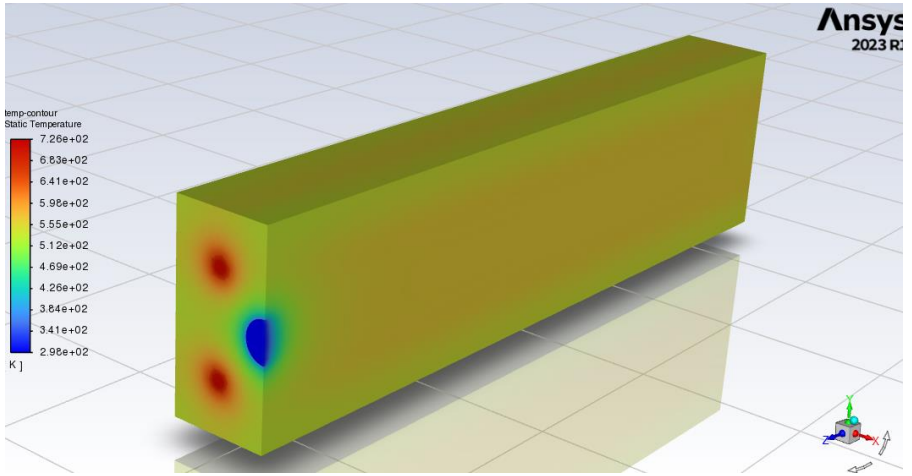
Temperature Contours – Mid-plane

Post-Charge



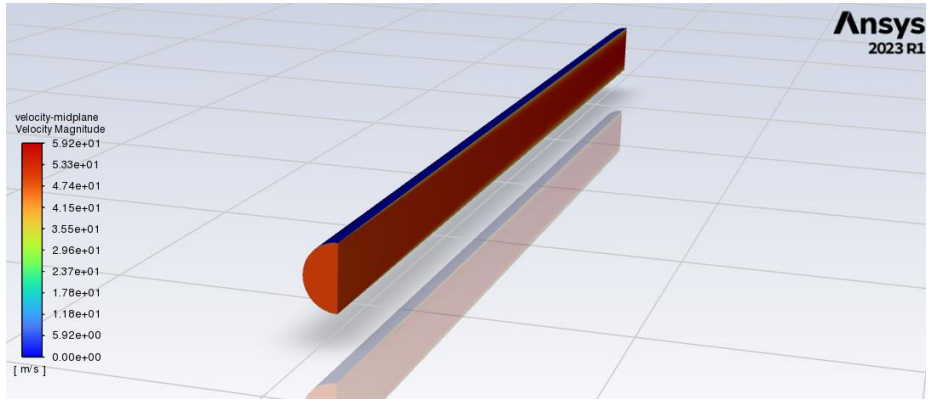


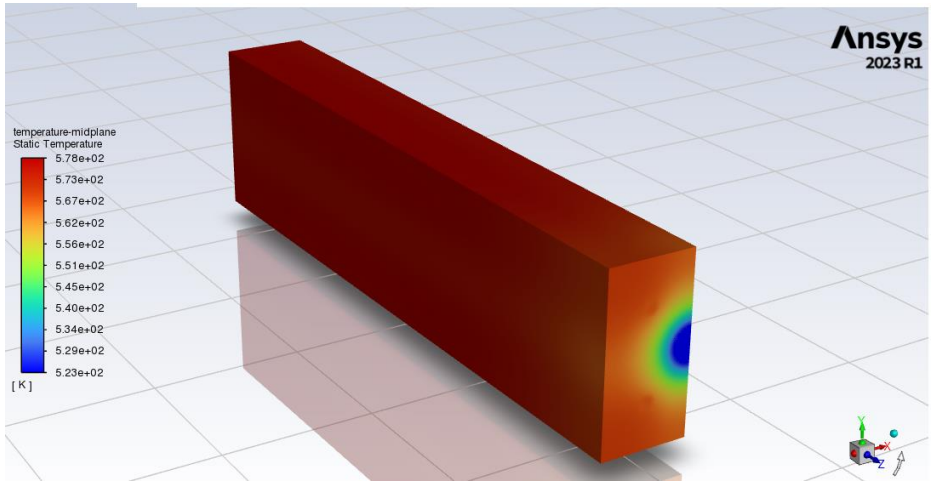




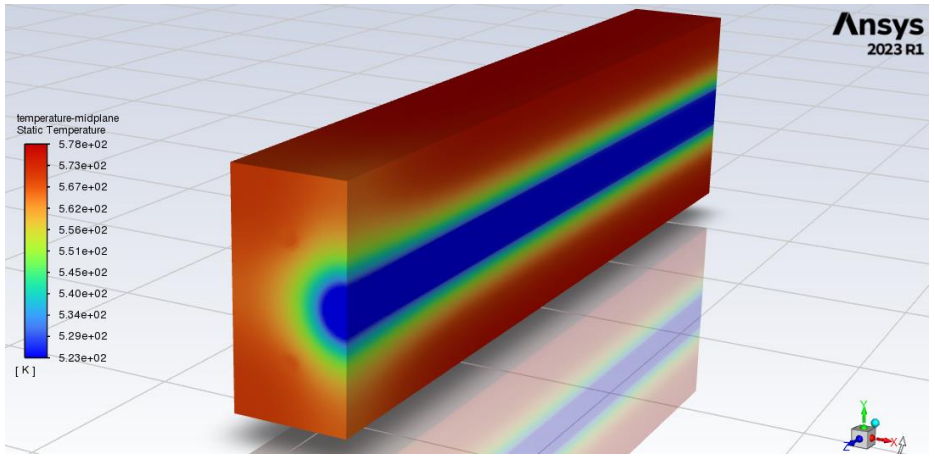
Thermal Contours

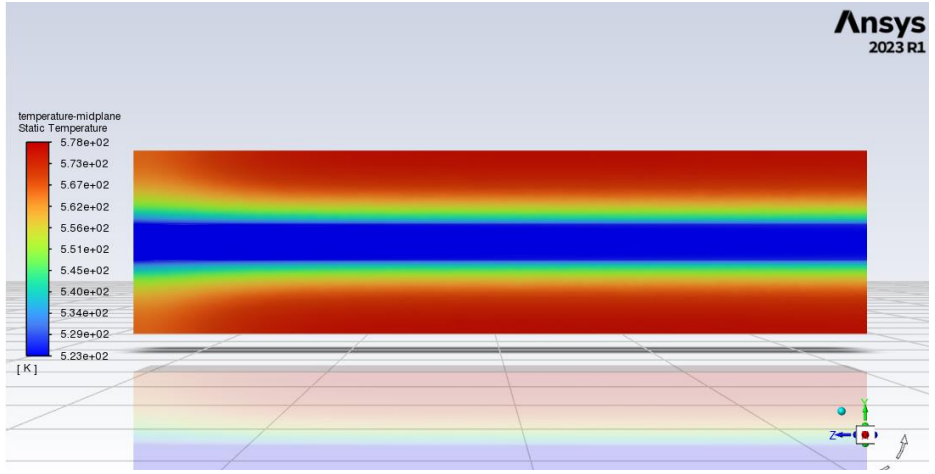
Velocity Field Post-Discharge





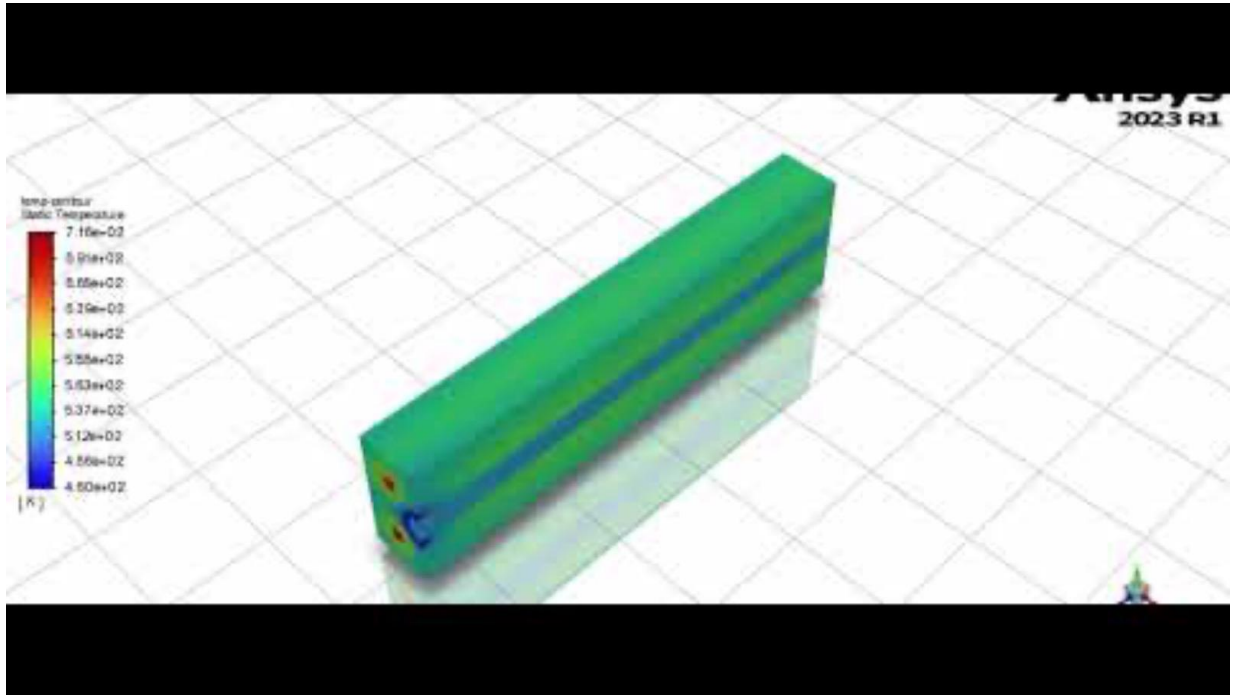
Midplane Temperature Post-Discharge





Temperature Evolution Animation

- Shows propagation of heat from heater into storage block during charge
- During discharge, cooler air gradually “eats” into the hot region near the tube
- Highlights slow conduction into corners due to low SM thermal conductivity (k = 0.4 W/m·K)



Energy & Efficiency Math

Initial temperature of SM: $T_1 = 298$ K

Volume-avg SM temperature at end of charge (t = 28,940 s): $T_2 = 576.4$ K

Volume-avg SM temperature at end of discharge (t = 35,960 s): $T_4 = 570.5$ K

SM properties: $\rho_{SM} = 1300$ kg/m³ $c_{p,SM} = 1200$ J/kg·K

Approximate SM volume (full block): $V_{SM} = 0.25 \times 0.125 \times 1 = 0.03125$ m³

Given Data

Initial temperature of SM: $T_1 = 298$ K

Volume-avg SM temperature at end of charge (t = 28,940 s): $T_2 = 576.4$ K

Volume-avg SM temperature at end of discharge (t = 35,960 s): $T_4 = 570.5$ K

SM properties: $\rho_{SM} = 1300$ kg/m³ $c_{p,SM} = 1200$ J/kg·K

Approximate SM volume (full block): $V_{SM} = 0.25 \times 0.125 \times 1 = 0.03125$ m³

Step 1 – Mass of storage material

$m_{SM} = \rho_{SM} \times V_{SM} = 1300 \times 0.03125 \approx 40.6$ kg

Step 2 – Total energy stored during charge

$E_{stored} = m_{SM} \times c_{p,SM} \times (T_2 - T_1)$

$T_2 - T_1 = 576.4 - 298 = 278.4$ K

$E_{stored} = 40.6 \times 1200 \times 278.4 \approx 1.36 \times 10^7$ J ≈ 13.6 MJ

Step 3 – Total energy removed during discharge

$E_{removed} = m_{SM} \times c_{p,SM} \times (T_2 - T_4)$

$T_2 - T_4 = 576.4 - 570.5 = 5.9$ K

$E_{removed} = 40.6 \times 1200 \times 5.9 \approx 2.89 \times 10^5$ J ≈ 0.29 MJ

Step 4 – Round-trip efficiency (based on storage material)

$\eta_{RT} = E_{removed} / E_{stored} = 0.29$ MJ / 13.6 MJ $\approx 0.021 \rightarrow 2.1\%$

Summary

Current setup: • Stores ~13.6 MJ in the block • Recovers only ~0.29 MJ during 2-hour discharge • Round-trip efficiency ≈ 2%

Slide bullet points: • Very low round-trip efficiency → discharge is too short and/or heat isn't reaching the air fast enough • Large portion of energy still trapped in corners far from tube (low k, short discharge time)

Optional: Energy based on HTF flow (formulas)

More formal energy expressions (similar to desalination-salt paper):

Charge: $E_{provided} = \int_{t_1}^{t_2} \dot{m}_{air} \times c_{p,air} \times [T_{in}(t) - T_{out}(t)]\, dt$

Discharge: $E_{recovered} = \int_{t_3}^{t_4} \dot{m}_{air} \times c_{p,air} \times [T_{out}(t) - T_{in}(t)]\, dt$

Overall efficiency: $\eta_{overall} = E_{recovered} / E_{provided}$

Note: The calculations above approximate these integrals using the McpΔT method with volume-averaged SM temperature.

Discussion

- Temperature rise in SM during charge matches expected linear heating from constant heater power.
- During 2-hour discharge, SM cools only ~6 K on average → most energy still stored.
- Likely reasons for low efficiency:
- Short discharge time; not enough time for heat to conduct from corners to tube
- Low SM thermal conductivity (k = 0.4 W/m·K) → slow diffusion
- High air flow rate → high outlet temperature early, but not enough total energy extracted

Conclusions & Future Work

- Built a 3D transient CFD model of an electro-thermal TES cell (charge + discharge).
- Generated required plots: SM temperature, HTF exit temperature, surface heat flux, and contour fields.
- With current settings:
- Stored ≈ **13.6 MJ** during 8-hour charge
- Removed ≈ **0.29 MJ** during 2-hour discharge
- Round-trip efficiency ≈ **2%** (based on SM energy)
- Future improvements:
- Longer discharge time and/or different flow rate
- Multiple tubes or higher-conductivity SM
- More realistic heat loss to ambient and comparison with experimental TES data.

Thank you